

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460272.6851 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.057 +/- 0.046
Transit depth (Rp/Rs)^2	0.33 +/- 0.53 [%]
Orbital Inclination (inc)	85.0 +/- 1.73
Airmass coefficient 1 (a1)	1.166 +/- 0.015
Airmass coefficient 2 (a2)	-0.1309 +/- 0.0067
Scatter in the residuals of the lightcurve fit is	1.28 %
Best Comparison Star	None
Optimal Aperture	4.79
Optimal Annulus	11.78
Transit Duration (day)	0.032 +/- 0.035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.057 ± 0.046 vs 0.131 (deviation 1.61σ)
Duration fit vs published	0.032 d vs 0.1075 d (deviation 2.16σ)
Mid-time O–C	8.7 min
Depth SNR	1.2
Residual scatter	1.28 %

Light curve

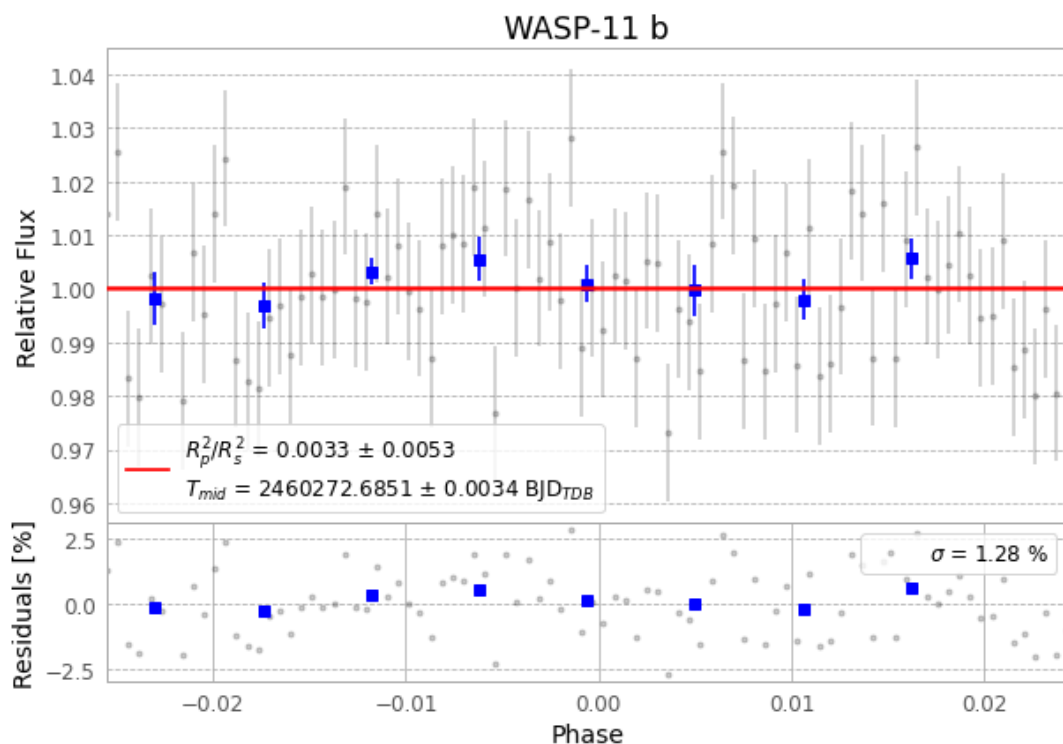


Figure 1 — FinalLightCurve_WASP-11 b_2023-11-23.png (SHA-256 e8f5ecba9c6f8126...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [371, 373], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 149646fdc3d39f70...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

90 FITS frames (59 MB), HATP-10231124020015.FITS → HATP-10231124062715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-231124.log (5abb78545816...), FinalLightCurve_WASP-11 b_2023-11-23.png (e8f5ecba9c6f...), FinalLightCurve_WASP-11 b_2023-11-23.csv (93fec1821111...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 03:49Z.